BundleSDF: Neural 6-DoF Tracking and 3D Reconstruction of Unknown Objects (CVPR 2023)

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Computer Vision Robotics Computer Graphics

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se(3)-TrackNet: Data-driven 6D Pose Tracking by Calibrating Image Residuals in Synthetic Domains B Wen, C Mitash, B Ren, K Bekris International Conference on Intelligent Robots and Systems (IROS)	182	2020
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Deep Learning Machine Learning Computer Graphics Rendering

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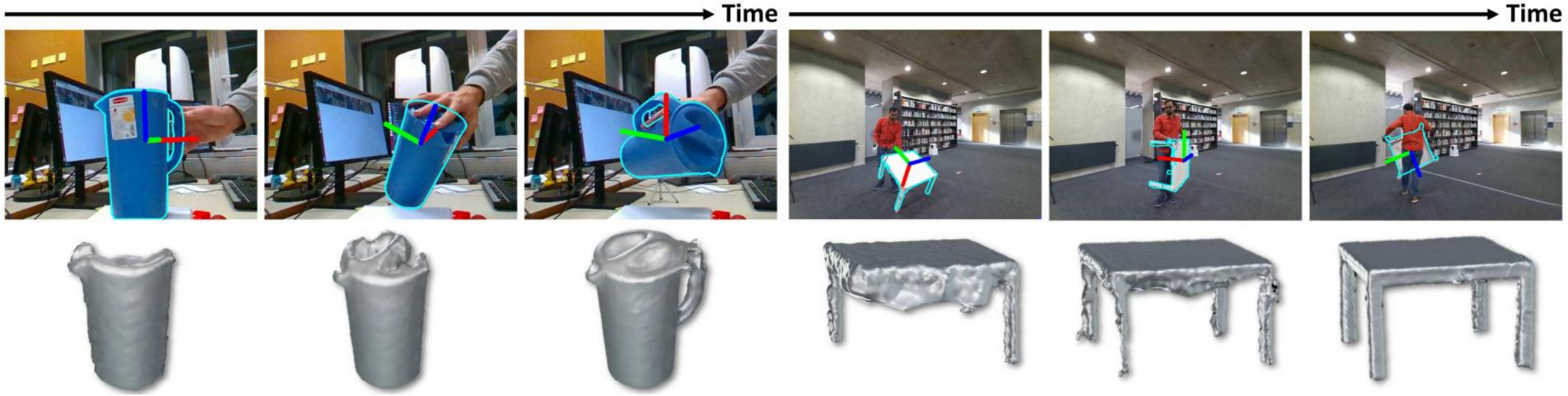
Posecnn: A convolutional neural network for 6d object pose estimation in cluttered scenes Y Xiang, T Schmidt, V Narayanan, D Fox arXiv preprint arXiv:1711.00199	2822	2017

Overview

BundleSDF:

method for **6-DoF tracking** of an **unknown object**,
while **simultaneously** performing **3D reconstruction** of the **object**.



Introduction

Background

Two fundamental (and closely related) problems in **computer vision**

- **3D reconstruction**
- **6-DoF pose tracking**

Why it matters?

- **Augmented Reality:** 새로운 객체를 즉시 등록하고 추적
- **Robotic Manipulation:** 처음 보는 물체를 잡고 조작
- **Learning from Demonstration:** 시연으로부터 학습
- **Sim-to-Real Transfer:** 실세계 데이터로 시뮬레이션 자산 생성

Introduction

Limitations of existing approaches

Prior efforts often **consider** these two problems **separately**

- **Neural Scene Representations**
  - GT 카메라 Pose와 Object Mask를 가정
  - Static Scene 위주, 객체의 모든 면을 보기 어려움 → **물체의 바닥**은 Reconstruction **X**
- **Object pose tracking**
  - Instance-Level: 대상 객체의 **CAD 모델** 필요
  - Category-Level: 학습된 카테고리 밖에서 **일반화 실패**

➔ Solve these two problems jointly

How…?

- similar in **spirit** to prior work in **object-level SLAM**

SLAM = Localization + Mapping
- **Localization ➔ Tracking**
- **Mapping ➔ Reconstruction**

Method

Architecture Overview

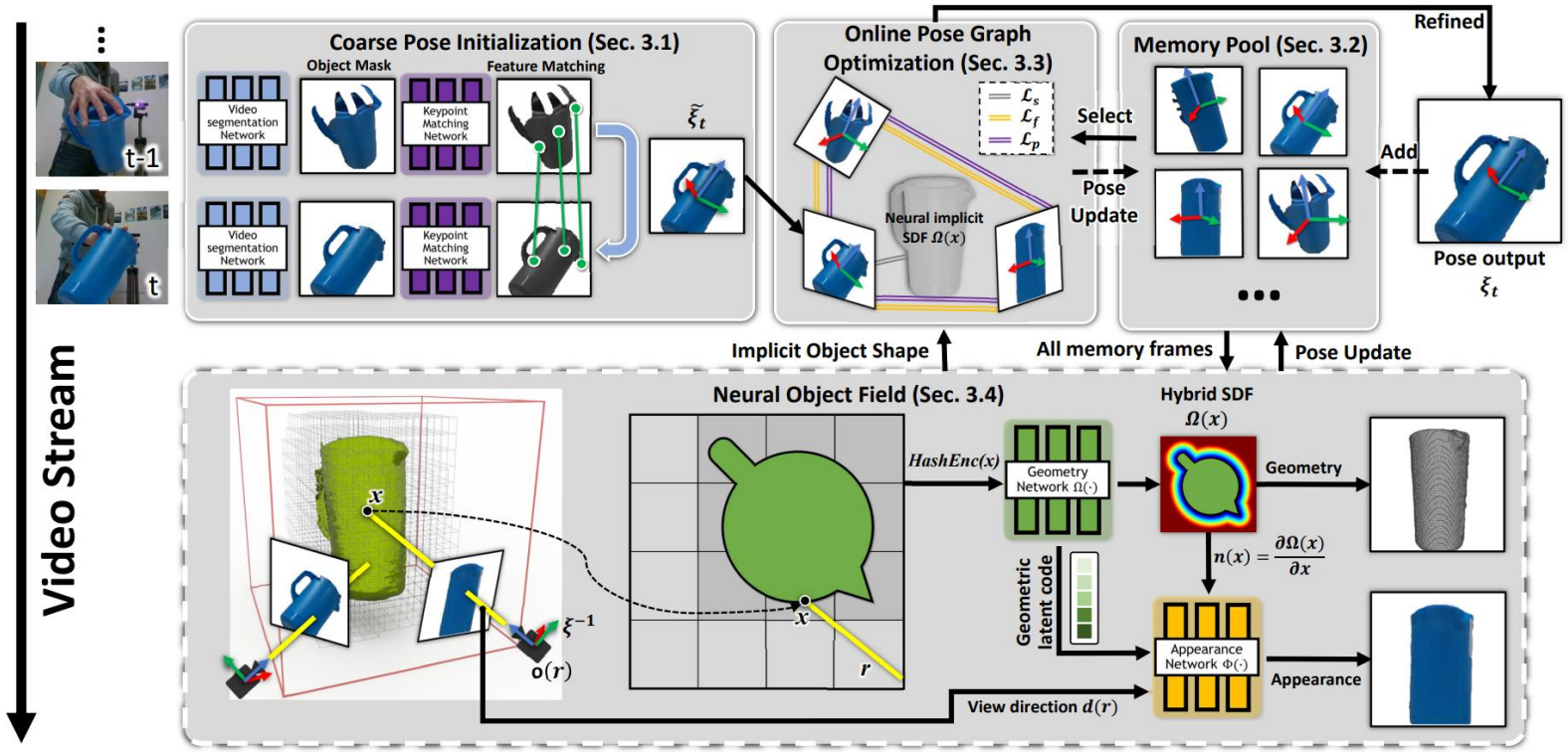
Input:

- RGB-D 비디오(Sequence)
- 첫 프레임의 객체 마스크

Output:

- 6-DoF OPE (프레임별)
- 3D Mesh

- **All processing** is **causal.**
- The **object** is assumed to be **rigid**.



Method

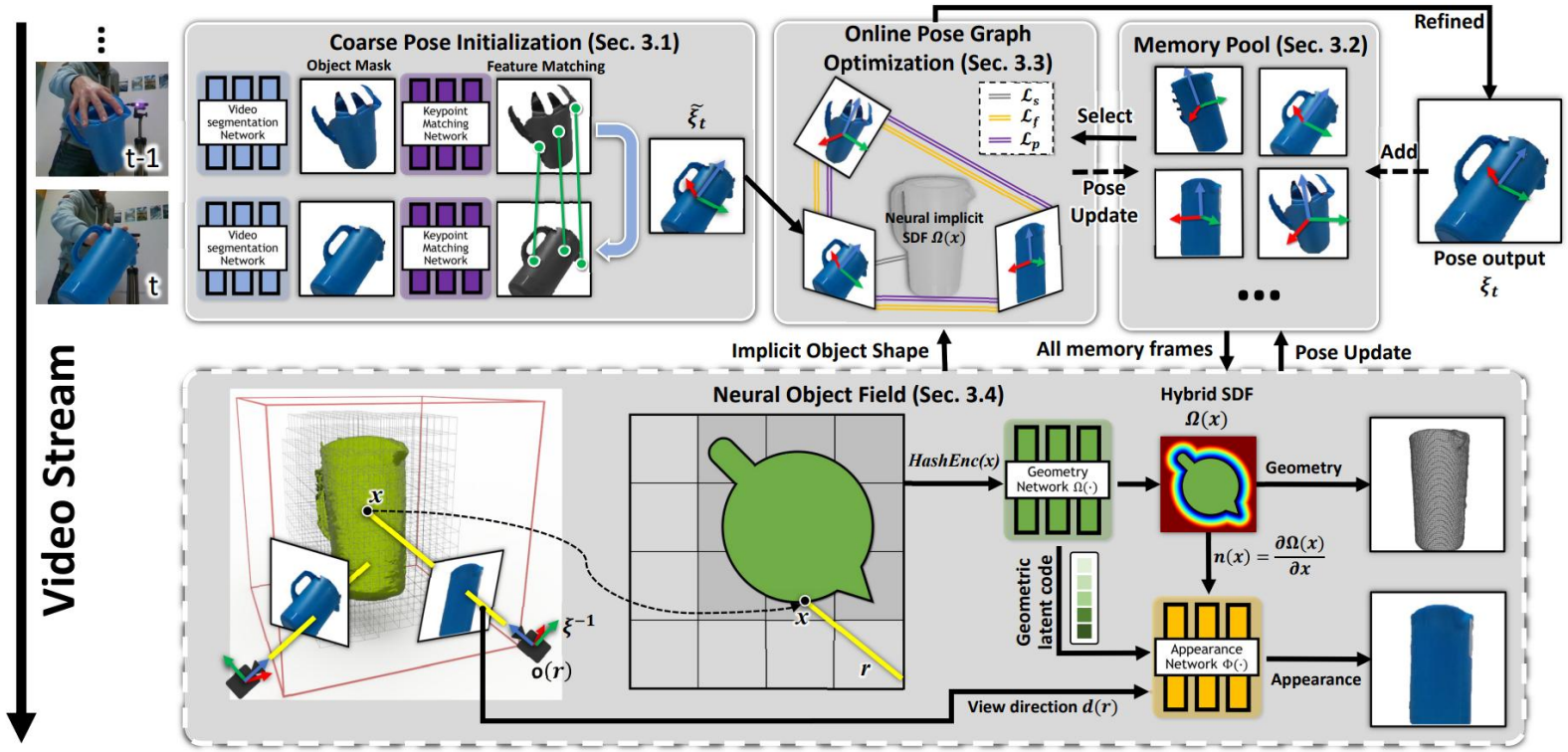
Architecture Overview

3.1. Coarse Pose Initialization

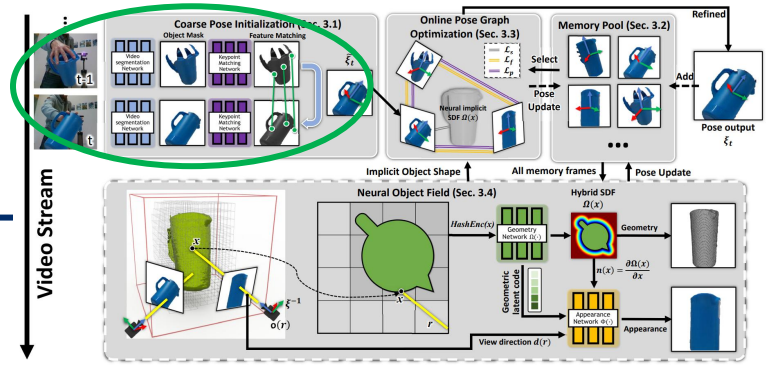
3.2. Memory Pool

3.3. Online Pose Graph Optimization

3.4. Neural Object Field



Method



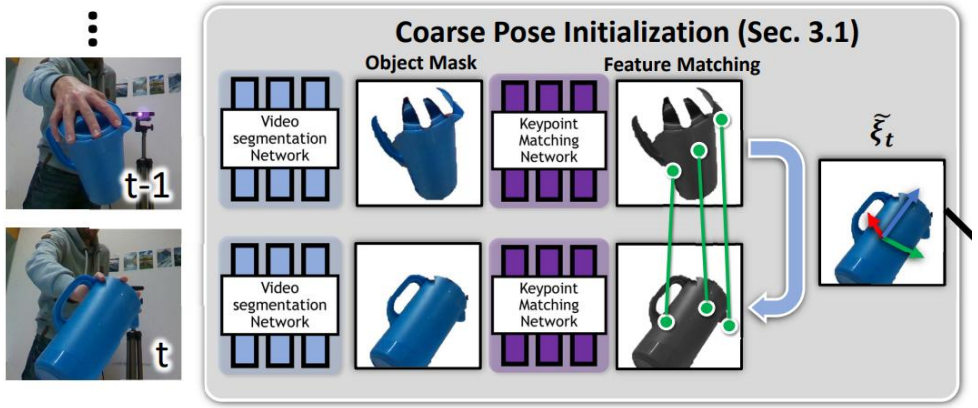
3.1. Coarse Pose Initialization

To provide a good **initial guess** for the subsequent online **pose graph optimization**.

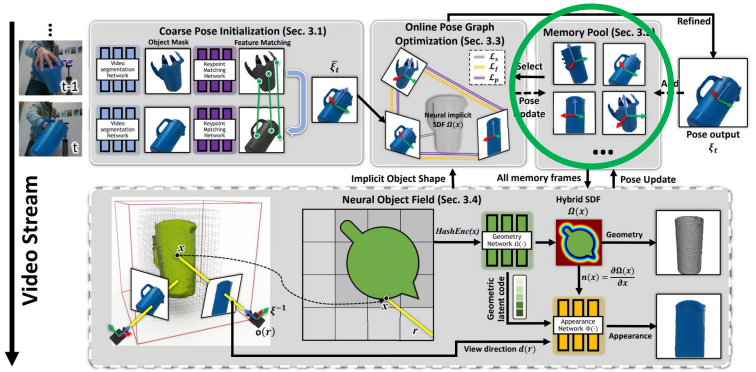
- $\tilde{\xi}_t \in \mathrm{SE}(3)$: coarse object pose estimate
- $\mathcal{F}_t$: current frame

1. The **object region** is **segmented** in $\mathcal{F}_t$ by leveraging a video **segmentation network** (XMem).
2. **Feature correspondences**(2D) in **RGB** between $\mathcal{F}_t$ and $\mathcal{F}_{t-1}$ are established via **LoFTR**.
3. Together **with depth**(2D → 3D), correspondences are **filtered by a RANSAC** using **Least Squares**.
4. The **pose hypothesis** that maximizes inliers is **selected** as $\tilde{\xi}_t$.

+ 첫번째 프레임에서의 물체의 좌표를 (0,0,0)으로 설정하고, 그것을 기준으로 계산



Method



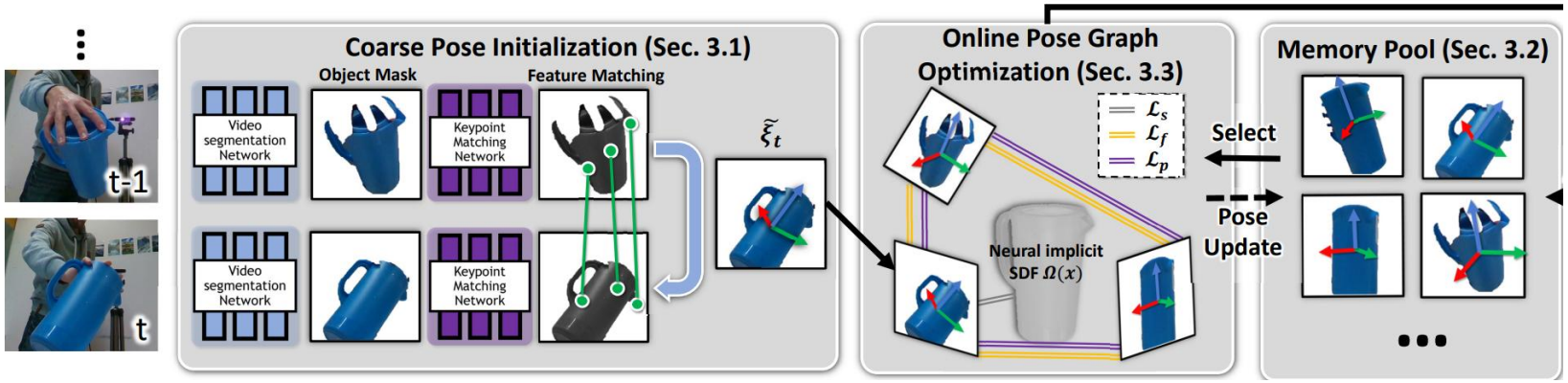
3.2. Memory Pool

It is important to **retain past information** to alleviate **long-term tracking drift**.

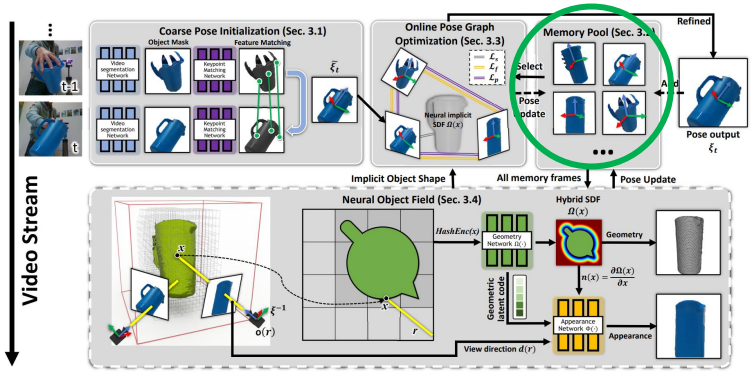
- introduce a **keyframe memory pool P**

Building Memory Pool

1. The **first frame F0** is **automatically added**, setting the **canonical coordinate system** for the **object**.
2. For **each new frame**, $\tilde{\xi}_t$ (coarse pose) is updated to ξ_t by **Online Pose Graph Optimization.** (in Sec. 3.3)
3. The frame is **only added** to **P** when its **viewpoint** is sufficient to **enrich multi-view diversity** in **P.**



Method



3.2. Memory Pool

Building Memory Pool

1. The **first frame F0** is **automatically added**, setting the **canonical coordinate system** for the **object**.
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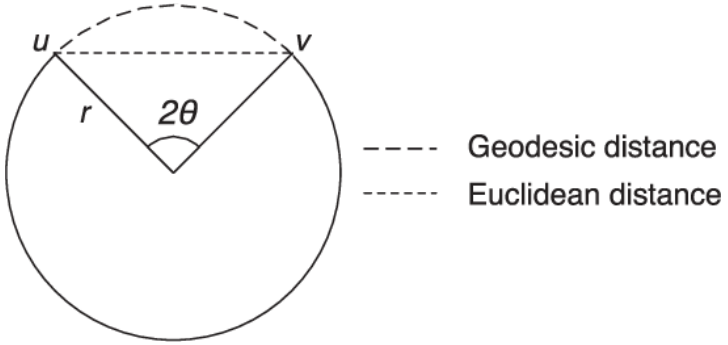
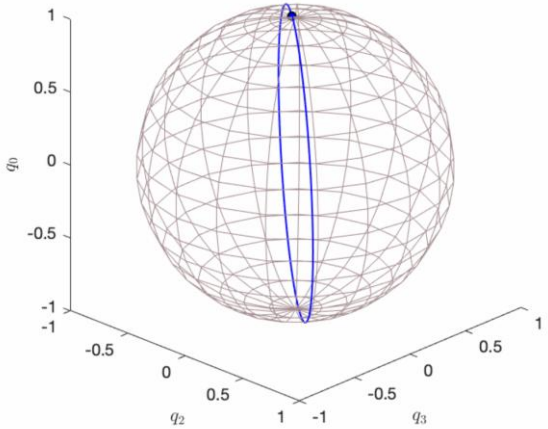
3. <u>The frame is **only added** to **P** when its **viewpoint** is sufficient to **enrich multi-view diversity** in **P.**</u>
   3.1. ξ_t is **compared** with **all poses** of memory frames in the **pool**.
   3.2. Since **in-plane object rotation** does **not** provide **additional information**,
   this **comparison** takes into account <u>**rotational geodesic distance**</u>**.**

In-Plane rotation





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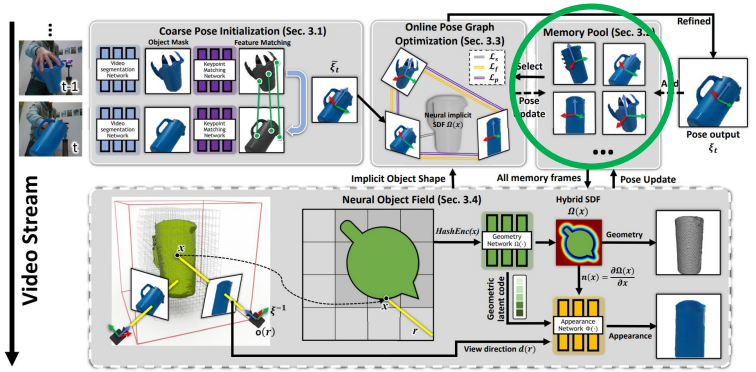


3.2. Memory Pool

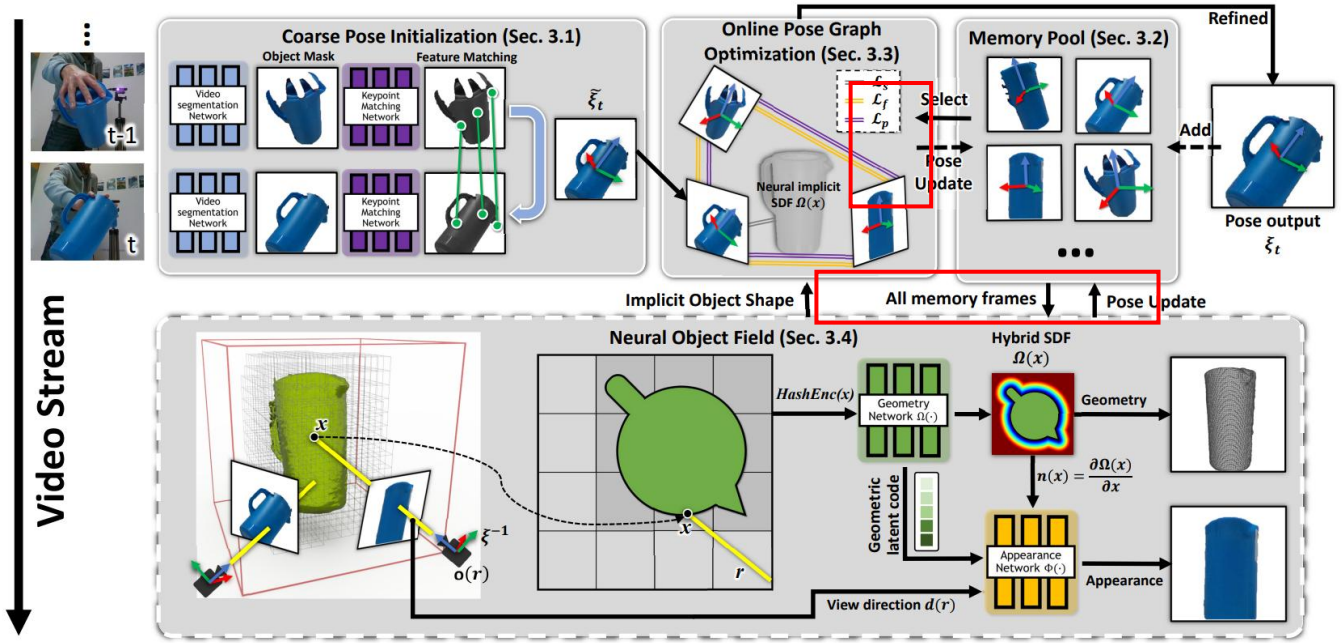
Pose Optimization

- **Online Pose Graph Optimization** (Sec. 3.3)
- **Neural Object Field** (Sec. 3.4)

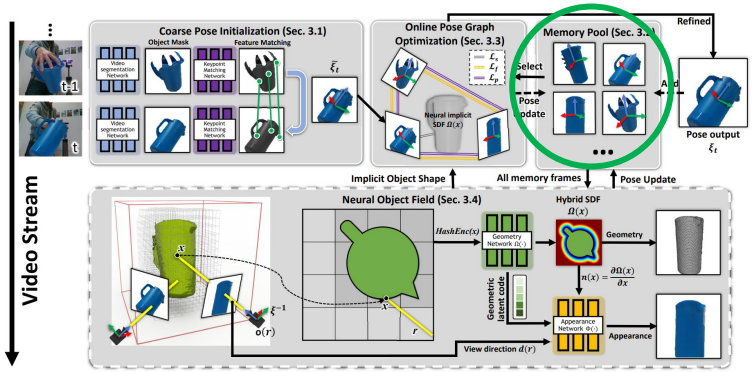
➔ Memory Pool을 통해 정보 교환

But, **같은** 프레임을 **동시에 수정**하면 **충돌**

Memory Pool이 NOF에게 Frame을 주고,
NOF에서 업데이트된 포즈를 돌려받음



Method



3.2. Memory Pool

Pose Optimization

- **Online Pose Graph Optimization** (Sec. 3.3)
- **Neural Object Field** (Sec. 3.4)

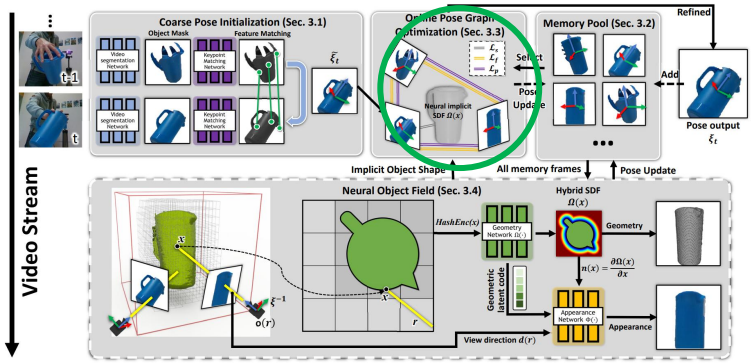
b(F) flag

Every frame in the memory pool has **a binary flag b(F)** and initializes as **FALSE.**

When updated by the **Neural Object Field → b(F) == True**

- **b(F) == False:** Neural Object Field 및 OPGO **모두 Pose** 업데이트 가능
- **b(F) == True :** Neural Object Field 만 **Pose** 업데이트 가능

Method



3.3. Online Pose Graph Optimization

새로운 Frame이 들어오면, **Memory Pool**에서 **K**개의 Frame을 **선택**해서 **OPGO** 진행.

Selecting Subset of Memory Frames

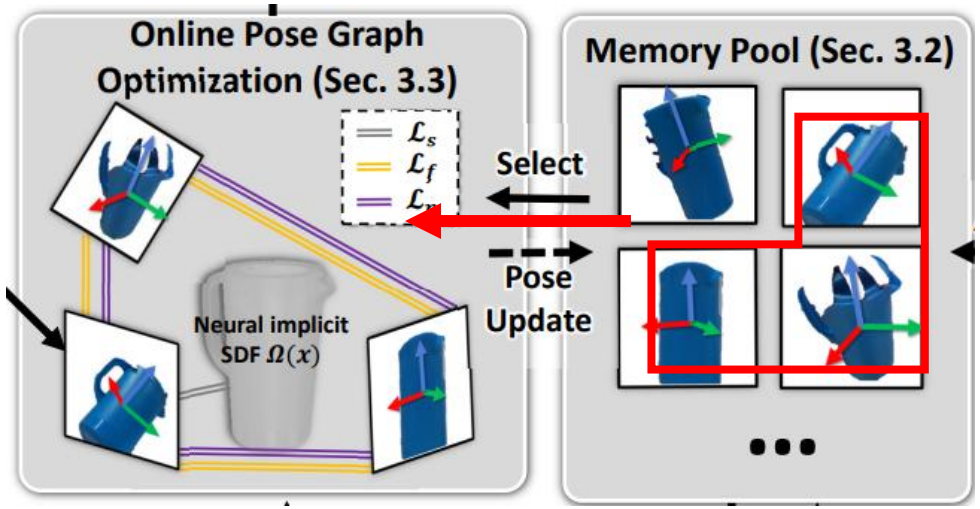
효율성을 위해서, optimization에 참가하는 frame을 **K개 이하**로 제한.

When |P| <= K,

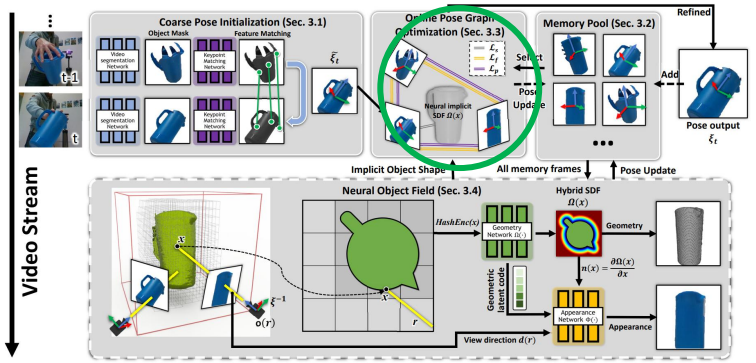
- **All frames** in the memory pool are **used**

When |P| > K,

- **selection process** is applied



Method

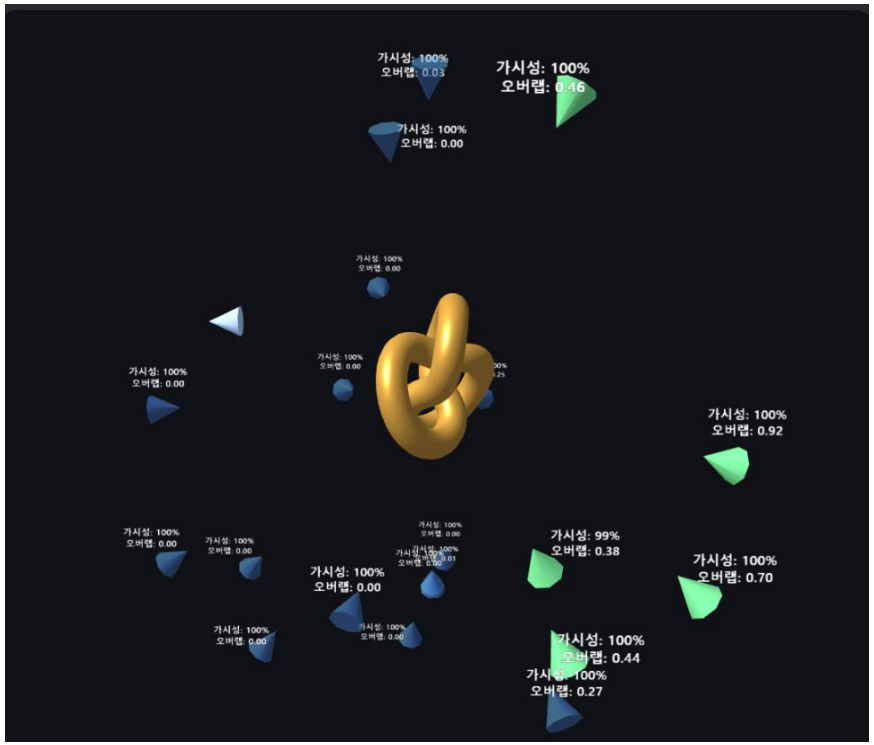


3.3. Online Pose Graph Optimization

Selection Process

1. Compute the **point normal map** for each frame $\mathcal{F}^{(k)}$ in the memory pool
2. Compute **visibility** (**normals @ ray direction** in the **new frame's** camera view)
3. If point cloud **visibility ratio** > **0.1**
4. measure the **viewing overlap** with $\mathcal{F}_t$ by computing the **rotation geodesic distance**
5. select the **K** memory **frames** with the **maximum viewing overlap**

→ **OPGO에 참가할 K개 프레임 :** $|\mathcal{P}_{pg}| = K$



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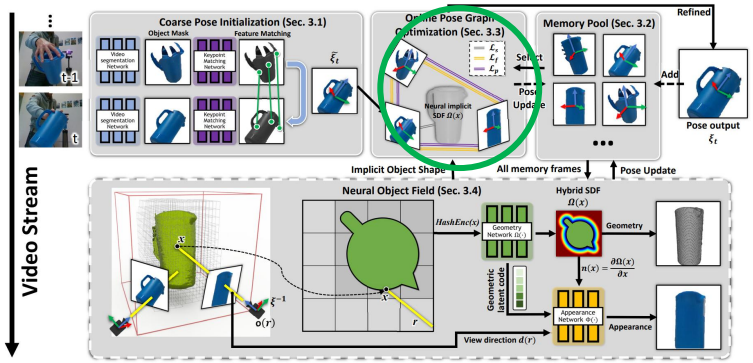


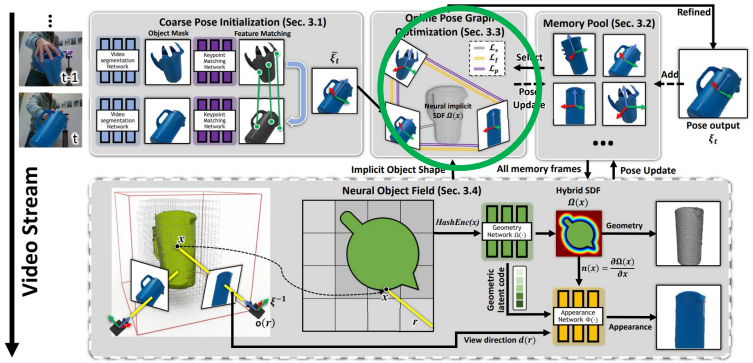
3.3. Online Pose Graph Optimization

Optimization

- The objective is to **find** the **optimal poses** that **minimize** the **total loss** of the pose graph
- $\mathcal{L}_p$, $\mathcal{L}_f$: pairwise edge losses
- $\mathcal{L}_s$: additional unary loss
- w_f, w_p, w_s : all set to 1 empirically
- $\mathcal{V} = \mathcal{F}_t \cup \mathcal{P}_{pg}$

$$\mathcal{L}_{pg} = w_s \mathcal{L}_s(t) + \sum_{i \in \mathcal{V}, j \in \mathcal{V}, i \neq j} \left[w_f \mathcal{L}_f(i,j) + w_p \mathcal{L}_p(i,j) \right], \quad (1)$$

Method



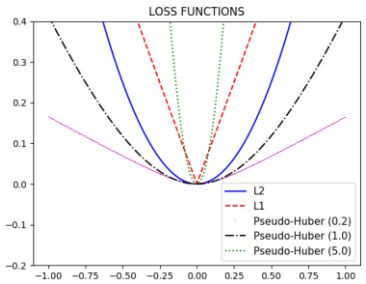
3.3. Online Pose Graph Optimization

Euclidean distance of the **RGBD**(3D) feature correspondences $p_m, p_n \in \mathbb{R}^3$

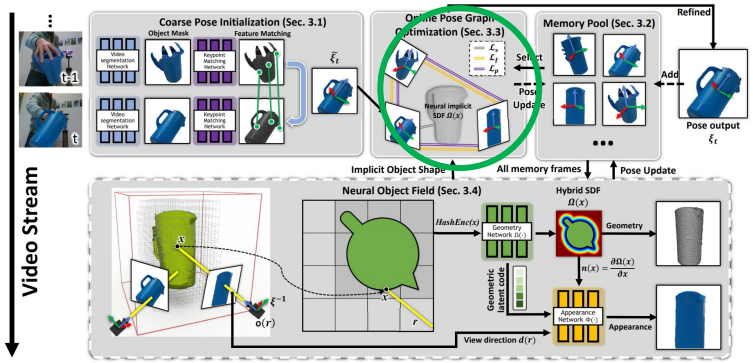
- ξ_i : object pose in frame $\mathcal{F}^{(i)}$
- $C_{i,j}$: correspondences between frames $\mathcal{F}^{(i)}$ and $\mathcal{F}^{(j)}$ via **LoFTR**

$$\mathcal{L}_f(i,j) = \sum_{(p_m,p_n)\in C_{i,j}} \rho\left(\left\|\xi_i^{-1}p_m - \xi_j^{-1}p_n\right\|_2\right) \qquad (2)$$

Huber loss



Method



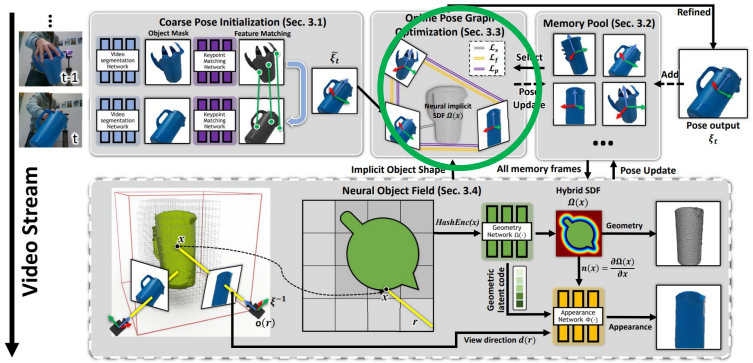
3.3. Online Pose Graph Optimization

Pixel-wise point-to-plane distance via **reprojective** association

- $T_{ij} \equiv \xi_j \xi_i^{-1}$: transforms from $\mathcal{F}^{(i)}$ to $\mathcal{F}^{(j)}$
- π_j : perspective projection mapping onto image I_j
- $\pi_{D_j}^{-1}$: inverse projection mapping via looking-up the depth image Dj at the pixel location
- $n_i(p)$: normal via looking-up the normal map of $\mathcal{F}^{(i)}$

$$\mathcal{L}_p(i,j) = \sum_{p \in I_i} \rho \left(\left| n_i(p) \cdot \left(T_{ij}^{-1} \pi_{D_j}^{-1} (\pi_j (T_{ij} p)) - p \right) \right| \right) \quad (3)$$

Method



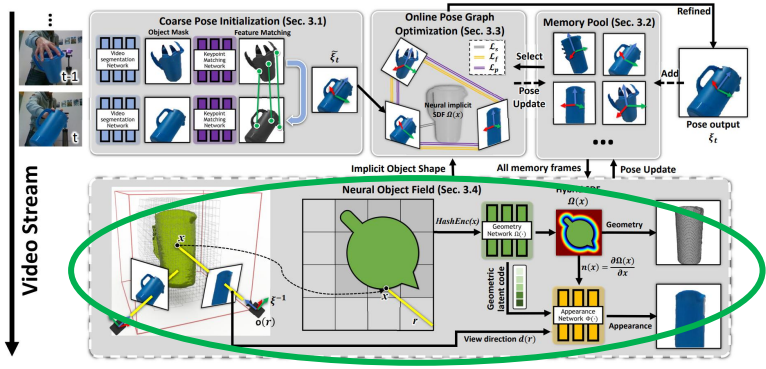
3.3. Online Pose Graph Optimization

Point-wise distance to the **neural implicit shape(SDF)**

- **only after** the initial **training** of the Neural Object Field **has converged.**
- $\Omega(\cdot)$: signed distance function (in Sec 3.4)

$$\mathcal{L}_s(t) = \sum_{p \in I_t} \rho\left(\left| \Omega(\xi_t^{-1}(\pi_D^{-1}(p))) \right| \right) \tag{4}$$

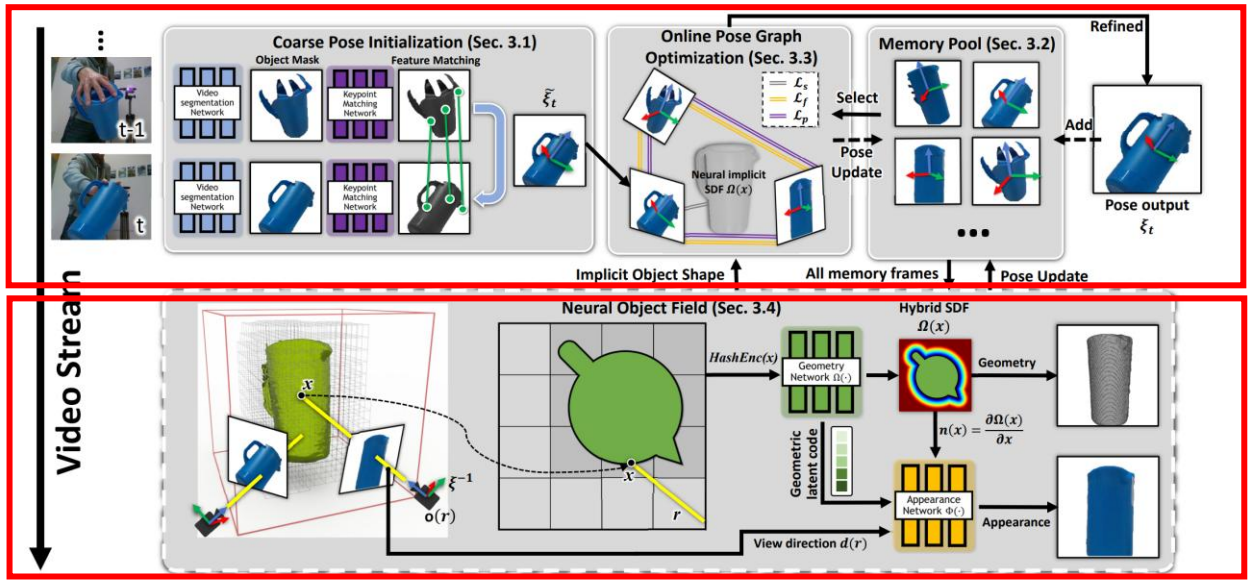
Method



3.4. Neural Object Field

Neural Object Field trains in a **separate thread parallel** to the online pose tracking.

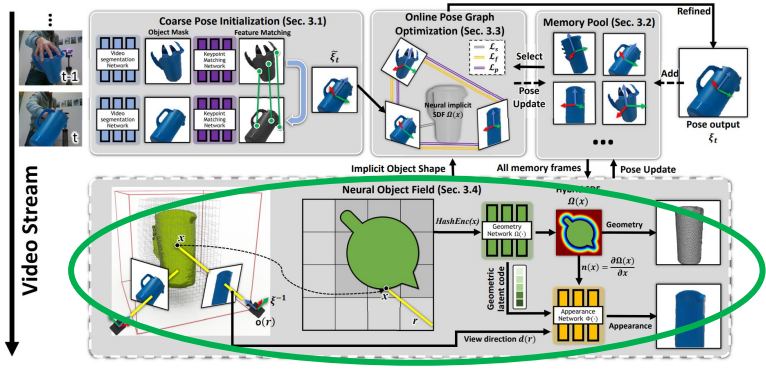
- At the start of **each training**, Neural Object Field **consumes all** memory pool's **frames** for **learning.**
- When training **converges**, **optimized poses** are **updated** to the **memory pool.**
- **Repeat**, whenever **new** memory **frames come in.**



➔Online Pose Tracking

➔Neural Object Field

Method

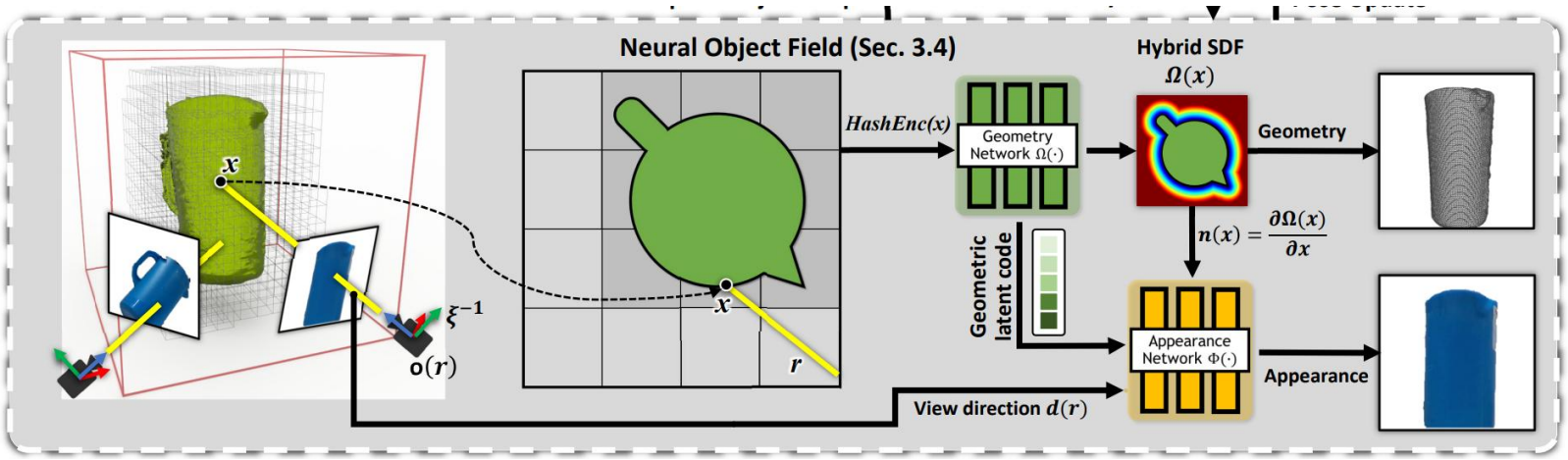


3.4. Neural Object Field

Field Representation.

- **Geometry function** $\Omega : x \mapsto s$
  - Input: 3D point $x \in \mathbb{R}^3$
  - Output: a signed distance value $s \in \mathbb{R}$

- **Appearance function** $\Phi : (f_{\Omega(x)}, n, d) \mapsto c$
  - Input: intermediate feature vector $f_{\Omega(x)}$, a point normal $n \in \mathbb{R}^3$, and a view direction $d \in \mathbb{R}^3$
  - Output: color $c \in \mathbb{R}^3_+$



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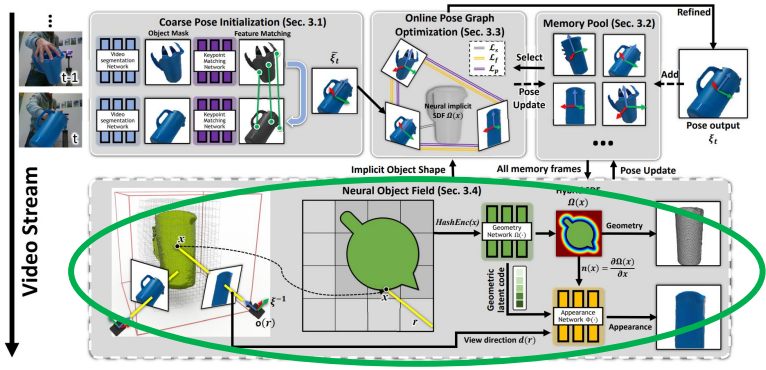


3.4. Neural Object Field

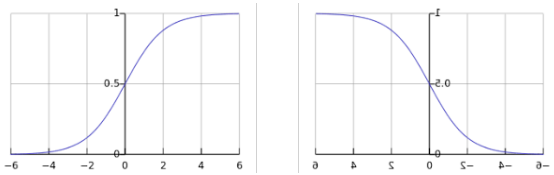
Rendering.

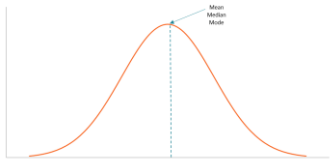
- color c of a ray r is integrated by near-surface regions

$$c(r) = \int_{z(r)-\lambda}^{z(r)+0.5\lambda} w(x_i)\underbrace{\Phi(f_{\Omega(x_i)}, n(x_i), d(x_i))}_{\text{샘플링된 점의 색상}}\, dt, \quad (6)$$

$$w(x_i) = \frac{1}{1+e^{-\alpha\Omega(x_i)}}\,\frac{1}{1+e^{\alpha\Omega(x_i)}}, \quad (7)$$

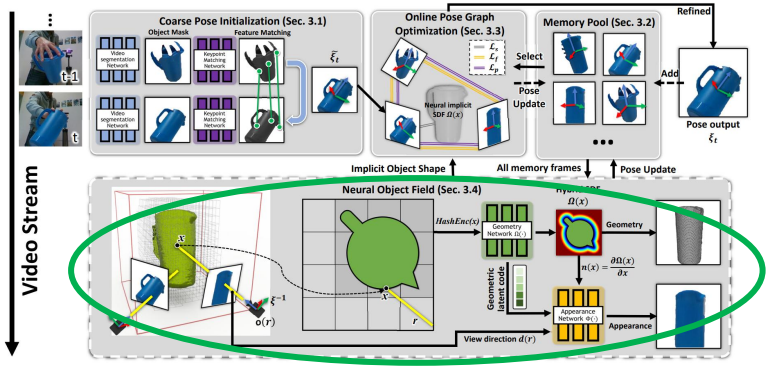
가중치





표면 근처에서 최댓값 → 표면에 가까울수록 색상 기여 큼

Method



3.4. Neural Object Field

Efficient Hierarchical Ray Sampling.

1. construct an **Octree representation** by **naively merging the point clouds** of the posed memory frames.
2. uniformly **sample N** points bounded by the **occupancy voxels** (gray boxes)

   (terminating at z(r) + 0.5λ)
3. Additional **N' samples** are allocated around the **surface**

➔ **Total N + N' samples**

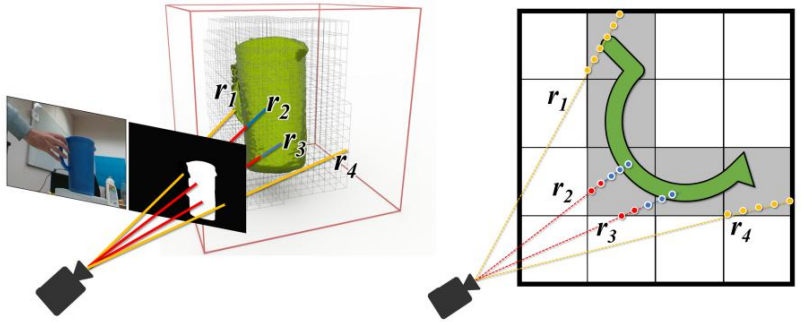
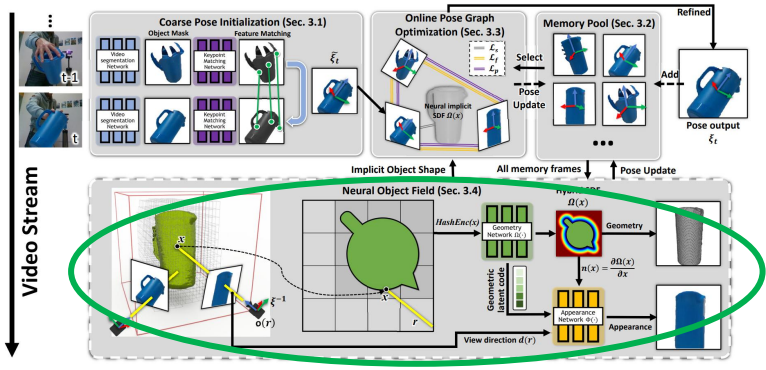


Figure 3. **Left:** Octree-voxel representation for efficient ray tracing, using the predicted binary mask from the video segmentation network (Sec. 3.1), which contains errors. Rays can land inside the mask (shown as red) or outside (yellow). **Right:** 2D top-down illustration of the neural volume and point sampling along the rays with hybrid SDF modeling. Blue samples are near the surface.

Method



3.4. Neural Object Field

Hybrid SDF Modeling.

1. Uncertain free space (yellow)
2. Empty space (red)
3. Near-surface space (blue)

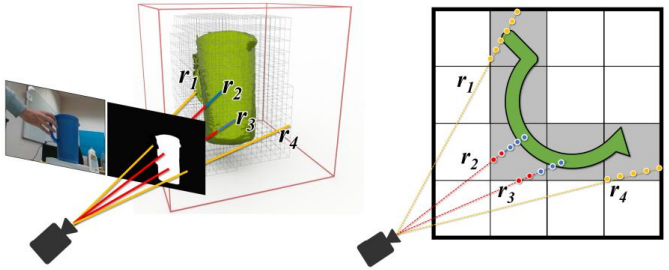


Figure 3. **Left:** Octree-voxel representation for efficient ray tracing, using the predicted binary mask from the video segmentation network (Sec. 3.1), which contains errors. Rays can land inside the mask (shown as red) or outside (yellow). **Right:** 2D top-down illustration of the neural volume and point sampling along the rays with hybrid SDF modeling. Blue samples are near the surface.

$$\mathcal{L}_u = \frac{1}{|\mathcal{X}_u|} \sum_{x \in \mathcal{X}_u} (\Omega(x) - \epsilon)^2. \tag{8}$$

➔ 불확실한 영역 = 일단 표면 바깥 (작은 양수 ε)으로 약하게 가정
(확신 X, 이후 신뢰할 만한 관측이 오면 쉽게 반영 가능하도록)

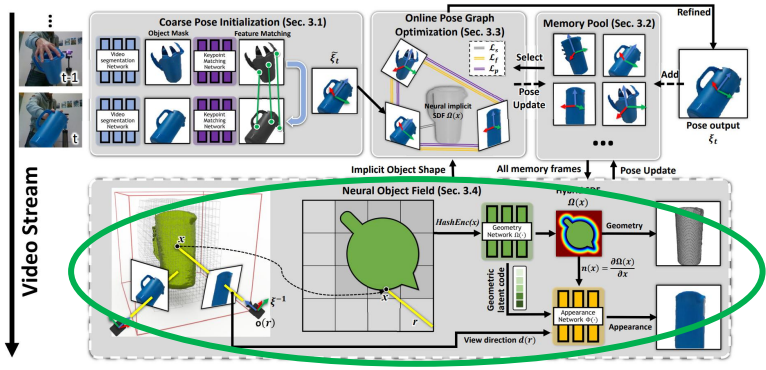
$$\mathcal{L}_e = \frac{1}{|\mathcal{X}_e|} \sum_{x \in \mathcal{X}_e} |\Omega(x) - \lambda|. \tag{9}$$

➔ 빈공간의 점과 표면과의 거리는 λ로 강제.
(표면과의 거리의 GT가 없으니, 최소한 λ이상 떨어져있다는 정보)

$$\mathcal{L}_{surf} = \frac{1}{|\mathcal{X}_{surf}|} \sum_{x \in \mathcal{X}_{surf}} \underbrace{(\Omega(x) + d_x - d_D)}_{\text{점과 표면과의 거리 + ray origin과 점의 거리= Depth}}{}^2, \tag{10}$$

➔ SDF가 정확한 표면 Depth를 예측하도록 Loss

Method



3.4. Neural Object Field

Total Loss.

$$\mathcal{L} = w_u\mathcal{L}_u + w_e\mathcal{L}_e + w_{surf}\mathcal{L}_{surf} + w_c\mathcal{L}_c + w_{eik}\mathcal{L}_{eik}, \qquad (11)$$

$$\mathcal{L}_c = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} \left\|\Phi(f_{\Omega(x)}, n(x), d(r)) - \bar{c}(r)\right\|_2, \qquad (12)$$

➔ GT랑 비교 L2 loss

$$\mathcal{L}_{eik} = \frac{1}{|\mathcal{X}_{surface}|}\sum_{x\in\mathcal{X}_{surface}} (\|\nabla\Omega(x)\|_2 - 1)^2. \qquad (13)$$

➔ 1만큼 움직이면 거리가 1만큼 변하도록
즉, 수직으로 이동 (gradient = 1)

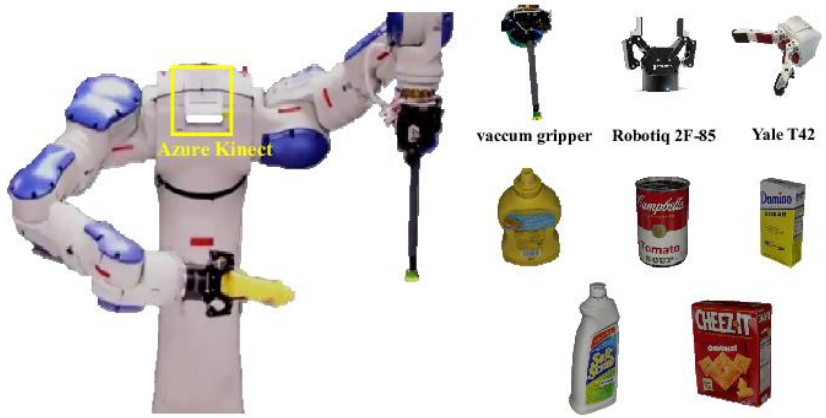
Experiment

Datasets

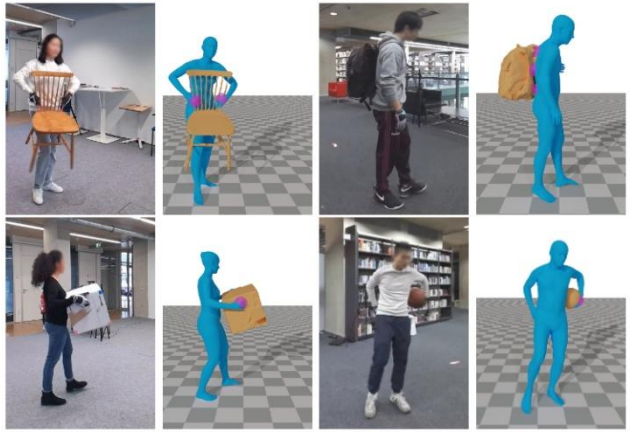
three real-world datasets

HO3D : RGBD video of a **human hand interacting** with YCB objects
YCBInEOAT : RGBD video of a **dual-arm robot manipulating** the YCB objects
BEHAVE : RGBD video of a **human body interacting** with the objects

HO3D



YCBInEOAT



BEHAVE

Experiment

Quantitative Comparison
(HO3D Dataset)

	Pose		Reconstruction
	ADD-S (%) ↑	ADD (%) ↑	CD (cm) ↓
NICE-SLAM [85]	22.29	8.97	52.57
SDF-2-SDF [53]	35.88	16.08	9.65
KinectFusion [43]	25.81	16.54	15.49
DROID-SLAM [61]	64.64	33.36	30.84
BundleTrack [69]	92.39	66.01	52.05
Ours	**96.52**	**92.62**	**0.57**

Table 1. Comparison on HO3D Dataset. ADD and ADD-S are AUC percentage (0 to 0.1 m). Reconstruction is measured by chamfer distance.

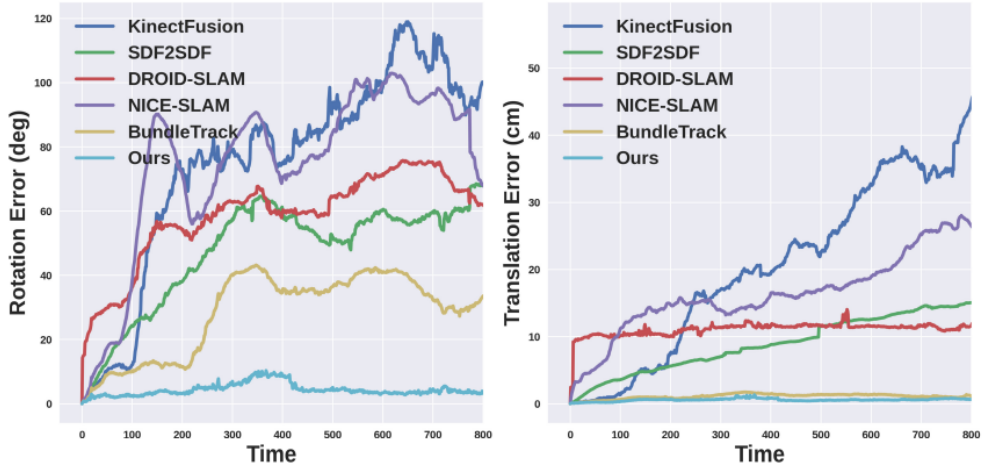


Figure 5. Pose tracking error against time on HO3D Dataset. Each time stamp's result is averaged across all videos. **Left:** Rotation error measured by geodesic distance. **Right:** Translation error.

Experiment

Qualitative Comparison
(HO3D Dataset)

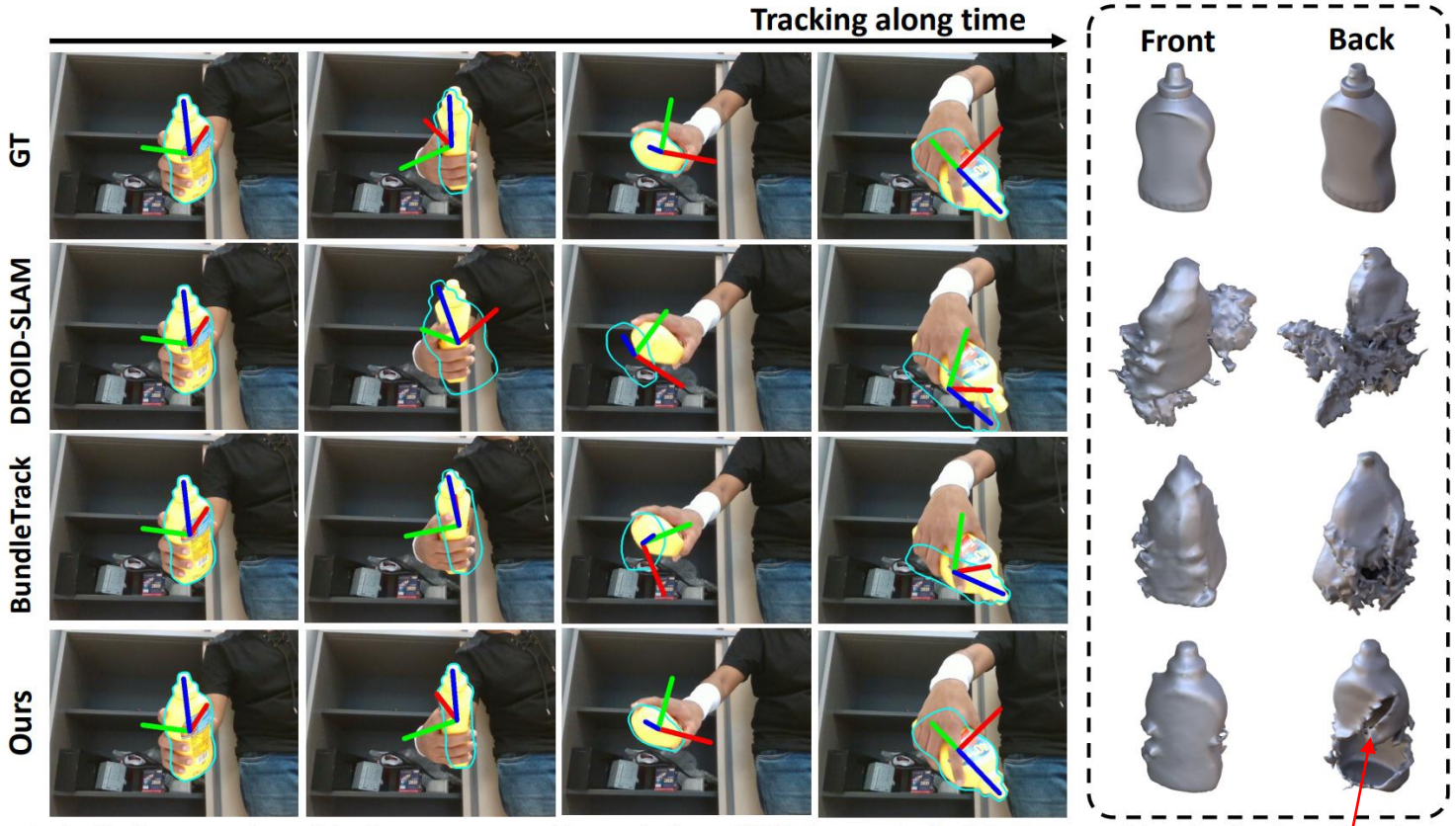


Figure 4. Qualitative comparison of the three most competitive methods on HO3D Dataset. **Left:** 6-DoF pose tracking visualization, where the contour (cyan) is rendered with the estimated pose. Note, as shown in the 2nd column, that our predicted pose sometimes corrects errors in the ground truth. **Right:** Front and back view of the final reconstructed shape output by each method. Due to hand occlusions, some parts of the object are never visible in the video. Meshes are rendered from the same viewpoint, though significant drift of DROID-SLAM and BundleTrack results in erroneously rotated meshes.

Experiment

Quantitative Comparison
(YCBInEOAT Dataset)

	Pose		Reconstruction
	ADD-S (%) ↑	ADD (%) ↑	CD (cm) ↓
NICE-SLAM [85]	23.41	12.70	6.13
SDF-2-SDF [53]	28.20	14.04	2.61
KinectFusion [43]	46.39	34.68	4.63
DROID-SLAM [61]	32.12	20.39	2.34
BundleTrack [69]	93.01	87.26	2.81
BundleTrack* [69]	92.53	**87.34**	-
MaskFusion* [50]	41.88	35.07	-
TEASER++* [78]	81.17	57.91	-
Ours	**93.77**	86.95	**1.16**

Table 2. Comparison on YCBInEOAT Dataset. ADD and ADD-S are AUC percentage (0 to 0.1 m). Reconstruction is measured by chamfer distance.

Experiment

Qualitative Comparison
(YCBInEOAT Dataset)

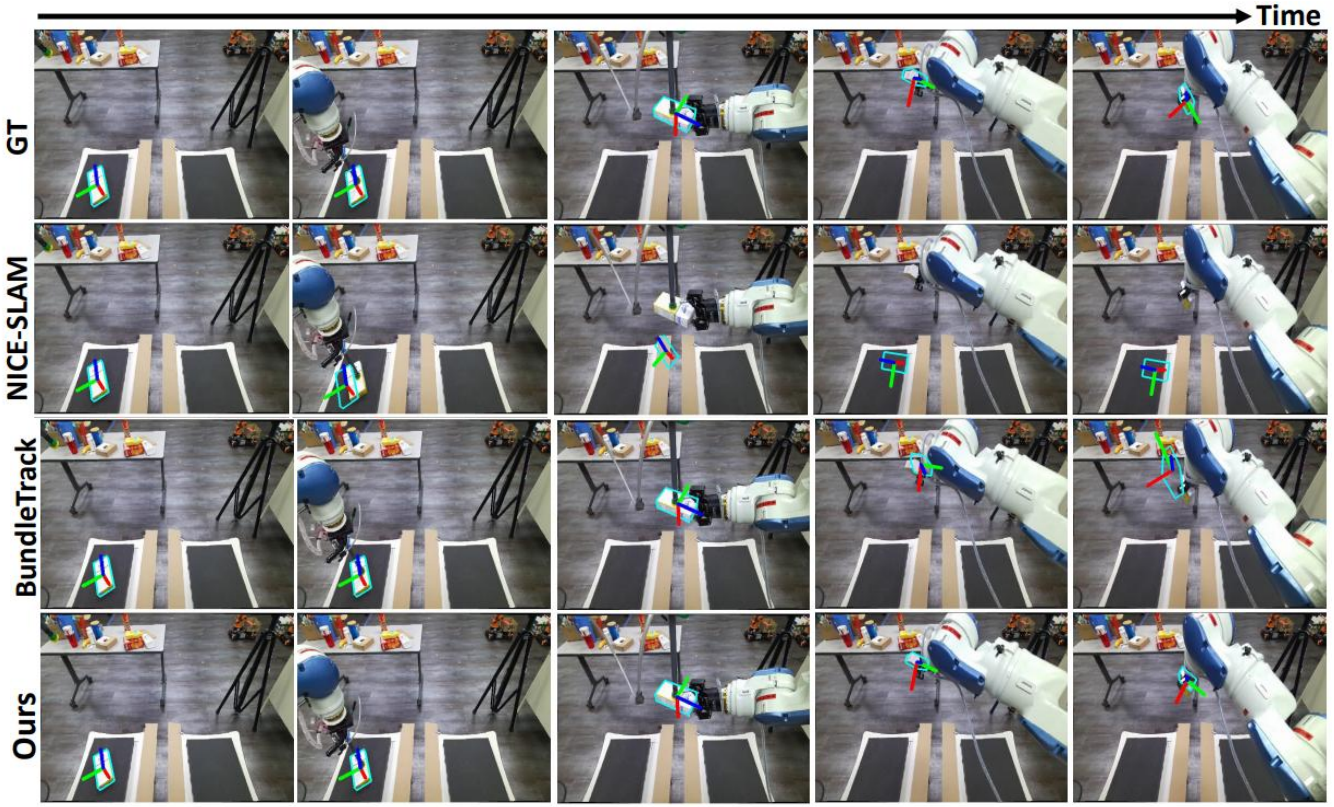


Figure 11. Qualitative comparison on YCBInEOAT video “sugar_box1".

Experiment

Quantitative Comparison
(BEHAVE Dataset)

	Pose		Reconstruction
	ADD-S (%) ↑	ADD (%) ↑	CD (cm) ↓
DROID-SLAM [61]	56.14	32.29	11.24
BundleTrack [69]	59.06	45.03	19.27
KinectFusion [43]	38.37	28.45	9.36
NICE-SLAM [85]	28.80	11.93	36.03
SDF-2-SDF [53]	25.71	10.05	35.99
Ours	83.63	67.52	4.66

세개의 Dataset 중 제일 어려운 Dataset

Table 3. Comparison on BEHAVE Dataset. ADD and ADD-S are AUC percentage (0 to 0.5 m). Reconstruction is measured by chamfer distance.

Experiment

Qualitative Comparison
(BEHAVE Dataset)

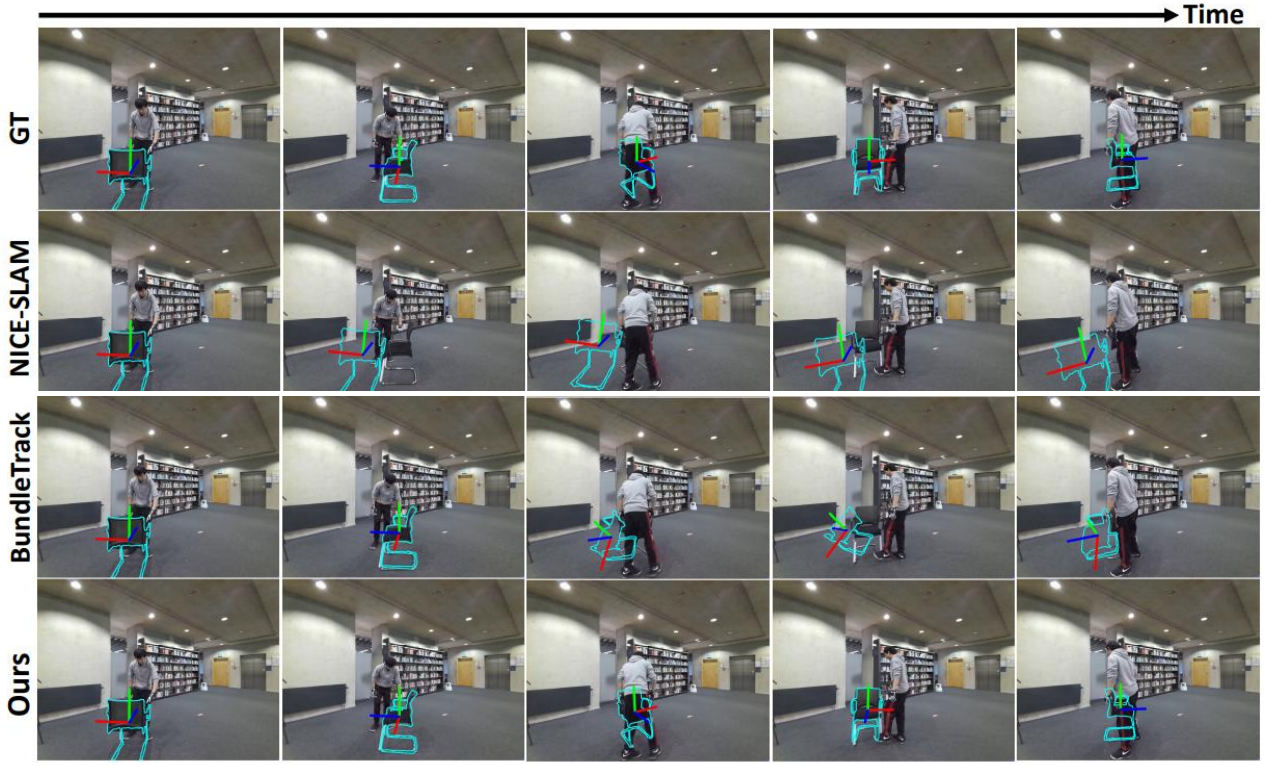


Figure 12. Qualitative comparison on BEHAVE video "Date03_Sub03_chairblack_hand.3". Our method is robust to severe and even complete occlusions (3rd and last column).

Experiment

Ablation Study

Ablations	Pose		Reconstruction
	ADD-S (%) ↑	ADD (%) ↑	CD (cm) ↓
Ours w/o memory	82.05	56.96	-
Ours w/o NOF	93.09	76.69	-
Ours-GPG	93.82	78.82	-
Ours w/o hybrid SDF	85.31	73.57	2.62
Ours w/o compact mem pool	87.48	59.99	0.90
Ours	**96.52**	**92.62**	**0.61**

Table 4. Ablation study of our design choices. *Ours w/o memory* removes the memory related modules and only performs frame-to-frame coarse pose estimation. *Ours w/o NOF* removes the Neural Object Field module and $\mathcal{L}_s$ in Eq. (1). *Ours-GPG* replaces the Neural Object Field by global pose graph optimization using all memory frames. It runs in a separate thread concurrently same as Neural Object Field. *Ours w/o hybrid SDF* only considers foreground rays in the mask instead of hybrid SDF modeling. *Ours w/o compact mem pool* adopts similar strategy of selecting frames to add into the memory pool as well as selecting the subset memory frames for pose graph optimization as in [69].

Experiment

Failure Mode

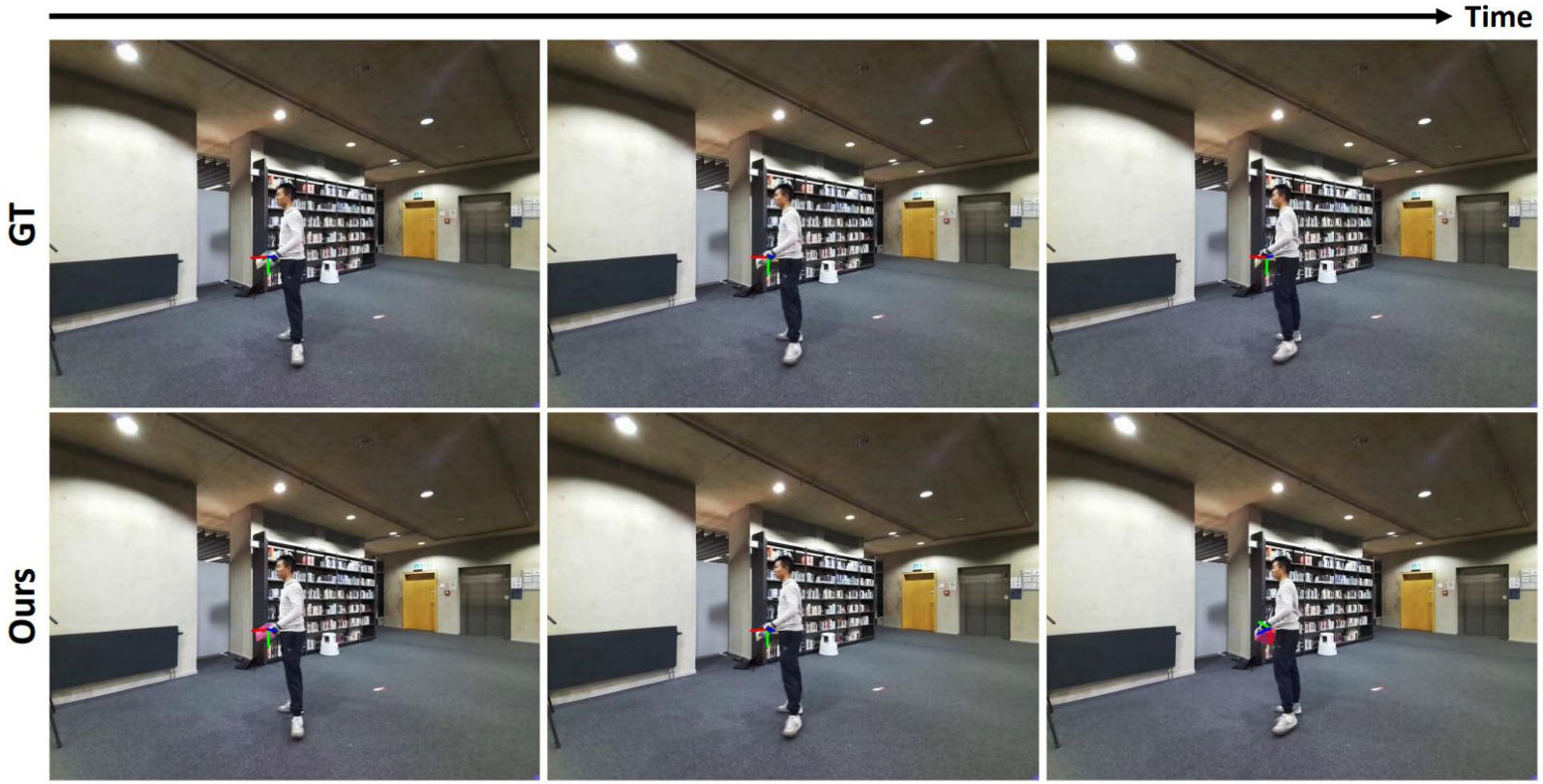


Figure 18. Failure case. The occurrence of severe occlusion, segmentation error, dearth of texture or geometric cues together lead to tracking failure. When the object re-appears, the recovered pose is affected by symmetric geometry.

Conclusion

Contribution

- Propose a **novel co-design** of **concurrent tracking** and neural **reconstruction** processes.
- Introduce a **hybrid SDF representation** to deal with **uncertain free space**.
- **state-of-the-art performance** on three public benchmarks.

Conclusion

Limitation

- **Not real-time** method.
- Tracking **fails** when **multiple** types of **challenges** appear **together**.
  (symmetric, transparent, occlusion, segmentation error)
- There is **no mechanism** to **auto-recover** when tracking fails.
- Method assumes the **object** to be **rigid**.